

SGA 6: Strict bilingual rebuild — Batch 068

Original source scan pages 92–110; Exposé I §§1.1.3 continued–2.10

1.1.3 (continued). If $\mathcal{C}$ and $\mathcal{C}'$ are triangulated S -categories and if F is an exact S -functor, then F defines a Cartesian S -functor

$$\mathrm{Tr}(\mathcal{C}) \longrightarrow \mathrm{Tr}(\mathcal{C}').$$

1.1.4. Let $\mathcal{C}$ be a triangulated S -category, and let $\mathcal{C}'$ be a full fibered sub- S -category of $\mathcal{C}$. We say that $\mathcal{C}'$ is a *triangulated sub- S -category* of $\mathcal{C}$ if, for every $X \in \mathrm{ob} S$, $\mathcal{C}'_X$ is a triangulated subcategory of $\mathcal{C}_X$ [V], Chap. I, §1, 2.3.

1.1.5. Let $\mathcal{C}$ be an additive S -category. One defines (SGA 4 XVII 2.2, 2.4) an additive (respectively triangulated) S -category

$$\mathrm{C}(\mathcal{C}) \quad \text{respectively} \quad \mathrm{K}(\mathcal{C}),$$

whose fiber over $X \in \mathrm{ob} S$ is the category $\mathrm{C}(\mathcal{C}_X)$ of complexes with differential of degree $+1$ (respectively the category $\mathrm{K}(\mathcal{C}_X)$ of complexes of $\mathcal{C}_X$ “up to homotopy”). If $\mathcal{C}$ is a flat abelian S -category, one defines, loc. cit., a triangulated S -category $\mathrm{D}(\mathcal{C})$, whose fiber over $X \in \mathrm{ob} S$ is the derived category $\mathrm{D}(\mathcal{C}_X)$; if $f : X \rightarrow Y$ is an arrow of S , we shall write

$$f^* : \mathrm{D}(\mathcal{C}_Y) \longrightarrow \mathrm{D}(\mathcal{C}_X)$$

for the inverse-image functor, obtained from

$$f^* : \mathrm{K}(\mathcal{C}_Y) \longrightarrow \mathrm{K}(\mathcal{C}_X)$$

by passage to the quotient. One defines full triangulated sub- S -categories

$$\mathrm{K}^*(\mathcal{C}) \quad \text{respectively} \quad \mathrm{D}^*(\mathcal{C}) \quad (* = +, -, b)$$

by the usual degree conditions.

1.2. Terminology

Let S be a site, let $\mathcal{C}$ be a flat abelian S -category (1.1), and let $\mathcal{C}_0$ be a strictly full fibered sub- S -category.

Let $F \in \mathrm{ob} \mathcal{C}_X$, for $X \in \mathrm{ob} S$. The set of arrows $f : U \rightarrow X$ in S/X such that there exists an epimorphism

$$E \longrightarrow f^*F, \quad E \in \mathrm{ob} \mathcal{C}_{0,U},$$

is a sieve on X . If this sieve is covering, we shall say that F is of $\mathcal{C}_0$ -finite type (or simply of finite type when no confusion is to be feared).

We shall say that $\mathcal{C}_0$ is *locally stable under kernels of epimorphisms* if the following condition is satisfied: for every $X \in \mathrm{ob} S$ and every epimorphism

$$u : E \longrightarrow F$$

of $\mathcal{C}_X$, with $E, F \in \mathrm{ob} \mathcal{C}_{0,X}$, the kernel of u is locally in $\mathcal{C}_0$; in other words the set of $f : U \rightarrow X$ in S/X such that

$$f^* \mathrm{Ker}(u) \in \mathrm{ob} \mathcal{C}_{0,U}$$

is a refinement of X .

By the standard process of decomposing a long exact sequence into short exact sequences, one sees that the preceding condition is equivalent to the following apparently stronger one: for every $X \in \text{ob } S$ and every exact sequence in $\mathcal{C}_X$

$$0 \longrightarrow E^0 \longrightarrow E^1 \longrightarrow \cdots \longrightarrow E^n \longrightarrow 0 \quad (*)$$

with $E^i \in \text{ob } \mathcal{C}_{0,X}$ for $1 \leq i \leq n$, the object E^0 is locally in $\mathcal{C}_0$.

We shall say that $\mathcal{C}_0$ is *locally quasi-stable under kernels of epimorphisms* if, for every exact sequence such as $(*)$, E^0 is of $\mathcal{C}_0$ -finite type.

On the other hand, we shall say that $\mathcal{C}_0$ is *locally liftable in $\mathcal{C}$* if, for every $X \in \text{ob } S$, every epimorphism

$$u : E \longrightarrow F$$

of $\mathcal{C}_X$ with $F \in \text{ob } \mathcal{C}_{0,X}$ locally admits a section. Equivalently, for $\mathcal{C}_0$ to be locally liftable in $\mathcal{C}$, it is necessary and sufficient that, for every diagram of $\mathcal{C}_X$, $X \in \text{ob } S$,

$$\begin{array}{ccc} & H & \\ & \downarrow & \\ E & \xrightarrow{u} & F \end{array}$$

where u is an epimorphism and $H \in \text{ob } \mathcal{C}_0$, the set of U over X for which there exists a commutative diagram

$$\begin{array}{ccc} & H_U & \\ \nearrow & \downarrow & \\ E_U & \xrightarrow{u_U} & F_U \end{array}$$

is a refinement of X .

We shall say that $\mathcal{C}_0$ is *locally quasi-liftable in $\mathcal{C}$* if the following weaker condition is satisfied: for every $X \in \text{ob } S$, and every epimorphism

$$u : E \longrightarrow F$$

of $\mathcal{C}_X$, with $F \in \text{ob } \mathcal{C}_{0,X}$, the set of arrows $f : U \rightarrow X$ of S/X for which there exists

$$v : F' \longrightarrow f^*E, \quad F' \in \text{ob } \mathcal{C}_{0,U},$$

such that $f^*(u)v$ is an epimorphism, is a refinement of X . This condition is equivalent to the following: for every diagram in $\mathcal{C}_X$

$$\begin{array}{ccc} & H & \\ & \downarrow & \\ E & \xrightarrow{u} & F, \end{array}$$

where u is an epimorphism and $H \in \text{ob } \mathcal{C}_0$, the set of U over X such that there exists a commutative diagram

$$\begin{array}{ccc} G & \xrightarrow{v} & H_U \\ \downarrow & & \downarrow \\ E_U & \xrightarrow{u_U} & F_U \end{array}$$

with $G \in \text{ob } \mathcal{C}_0$ and v an epimorphism, is a refinement of X .

If S is a chaotic site, for example the punctual site (one object, one morphism), we shall omit the adverb “locally” in the preceding definitions.

1.3. Examples

a) Let S be the punctual site, let A be a ring, and let $\mathcal{C}$ be the category of left A -modules. The full subcategory $\mathcal{C}_0$ of $\mathcal{C}$ formed by the projective A -modules of finite type is stable under kernels of epimorphisms and is liftable in $\mathcal{C}$; an A -module is of $\mathcal{C}_0$ -finite type if and only if it is of finite type in the ordinary sense. The full subcategory $\mathcal{C}_1$ of $\mathcal{C}$ formed by the A -modules of finite type is quasi-liftable in $\mathcal{C}$, but is not liftable in general; if A is noetherian, $\mathcal{C}_1$ is stable under kernels of epimorphisms. The full subcategory $\mathcal{C}'_0$ of $\mathcal{C}$ formed by the free A -modules of finite type is liftable in $\mathcal{C}$ and quasi-stable, though not stable in general, under kernels of epimorphisms.

b) Let $(S, \mathcal{O}_S)$ be a space ringed by local rings, let S be the site of open subsets of S , and let $\mathcal{C}$ be the fibered S -category, even split (SGA 1 VI 9), defined by the functor

$$U \longmapsto \{\text{category of left } \mathcal{O}_U\text{-modules}\}.$$

Then $\mathcal{C}$ is a flat abelian S -category (1.1). Let $\mathcal{C}_0$ be the strictly full additive sub- S -category of $\mathcal{C}$ defined by

$$U \longmapsto \{\text{category of finite-type free } \mathcal{O}_U\text{-modules}\}.$$

An $\mathcal{O}_S$ -module M is locally of $\mathcal{C}_0$ -finite type if and only if M is of finite type in the ordinary sense (EGA 0_I, 5.2). From EGA 0_I, 5.2.7 and 5.4.9, it follows that $\mathcal{C}_0$ is locally stable under kernels of epimorphisms and locally liftable in $\mathcal{C}$.

c) Let G be a group, let S be the topos of G -sets (SGA 4 V 3.11), and let k be a field, regarded as the constant ring on S . Let $\mathcal{C}$ be the split S -category defined by

$$U \longmapsto \{\text{category of left } k_U\text{-modules}\};$$

then $\mathcal{C}$ is a flat abelian S -category. Let $\mathcal{C}_0$ be the strictly full additive sub- S -category of $\mathcal{C}$ defined by

$$U \longmapsto \{\text{category of finite-type free } k_U\text{-modules}\}.$$

Since localization in S is equivalent to forgetting the operations of G , it is clear that a G - k -module M is locally of $\mathcal{C}_0$ -finite type if and only if M is finite-dimensional as a vector space over k , and that $\mathcal{C}_0$ is locally stable under kernels of epimorphisms and locally liftable in $\mathcal{C}$.

The reader is left to look at the ‘‘Galois’’ variant of c), where G is a profinite group and S is the topos of sets on which G operates continuously.

In fact, examples b) and c) are particular cases of the following example.

d) Let $(S, \mathcal{O}_S)$ be a ringed topos (cf. SGA 4 IV 2.10), and let $\mathcal{C}$ be the split S -category defined by

$$U \longmapsto \{\text{category of left } \mathcal{O}_U\text{-modules}\};$$

then $\mathcal{C}$ is a flat abelian S -category. Let $\mathcal{C}_0$ be the strictly full additive sub- S -category of $\mathcal{C}$ defined by

$$U \longmapsto \{\text{category of finite-type free } \mathcal{O}_U\text{-modules}\}.$$

An $\mathcal{O}_S$ -module locally of $\mathcal{C}_0$ -finite type will simply be called *of finite type*. The category $\mathcal{C}_0$ is locally quasi-stable under kernels of epimorphisms and locally liftable in $\mathcal{C}$, as follows from:

Lemma 1.3.1. Let P be an $\mathcal{O}_S$ -module. The following conditions are equivalent:

- (i) P is of finite type and the functor $\text{Hom}(P, \cdot)$ is exact;
- (ii) P is of finite type and every epimorphism $M \rightarrow P$ locally admits a section;
- (iii) P is locally a direct factor of a free module of finite type.

The proof is immediate and is left as an exercise to the reader.

1.4. An “extension” lemma (cf. EGA 0_{III}, 11.9.1)

Proposition 1.4.1. Let S be a site, let $\mathcal{C}$ be a flat abelian S -category (1.1), and let $\mathcal{C}_0$ be a strictly full additive sub- S -category of $\mathcal{C}$. Assume that $\mathcal{C}_0$ is locally quasi-liftable in $\mathcal{C}$ (1.2). Let $S \in \text{ob } S$, let

$$u : E \longrightarrow F$$

be a morphism, of degree 0, of complexes of $\mathcal{C}_S$ (differentials of degree +1), let G be the cone of u (cf. [V], Chap. I, §1, 2.1), let $n \in \mathbb{Z}$, and assume that $H^n(G)$ is of $\mathcal{C}_0$ -finite type (1.2). Then the set of $X \in \text{ob}(S/S)$ satisfying property (P) below is a refinement of S .

(P): there exist an object $M \in \mathcal{C}_{0,X}$ and arrows of $\mathcal{C}_X$

$$r : M \longrightarrow Z^{n+1}(E_X) = \text{Ker}\left(E_X^{n+1} \xrightarrow{d^{n+1}} E_X^{n+2}\right), \quad s : M \longrightarrow F_X^n$$

such that the following diagram of $\mathcal{C}_X$

$$\begin{array}{ccccccc} \cdots & \longrightarrow & E_X^{n-1} & \xrightarrow{(d_{E_0}^{n-1})} & E_X^n \oplus M & \xrightarrow{(d_E^n, r)} & E_X^{n+1} \longrightarrow \cdots \\ & & \downarrow u_X^{n-1} & & \downarrow (u_X^n, s) & & \downarrow u_X^{n+1} \\ \cdots & \longrightarrow & F_X^{n-1} & \xrightarrow{d_F^{n-1}} & F_X^n & \xrightarrow{d_F^n} & F_X^{n+1} \longrightarrow \cdots \end{array}$$

is a morphism of complexes $u' : E' \rightarrow F_X$, that is,

$$d_F^n s = u_X^{n+1} r,$$

and such that

(i) $H^{n+1}(u')$ is a monomorphism, with

$$\Im H^{n+1}(u') = \Im H^{n+1}(u_X);$$

(ii) $H^n(u')$ is an epimorphism.

In pictorial terms, one may say that, after modifying E and u locally in degree n only by adding, in degree n , an object of $\mathcal{C}_0$, one can locally make $H^{n+1}(u)$ injective, without changing its image, and $H^n(u)$ surjective.

Proof. The procedure is as follows: first one shows that r and s may be chosen locally so as to make $H^{n+1}(u)$ injective; one verifies that the finiteness hypothesis on $H^n(G)$ is preserved; a new local correction in degree n then makes it possible to obtain $H^n(u)$ surjective.

The exact cohomology sequence

$$\cdots \longrightarrow H^n(G) \longrightarrow H^{n+1}(E) \longrightarrow H^{n+1}(F) \longrightarrow \cdots$$

shows that the kernel of $H^{n+1}(u)$ is of $\mathcal{C}_0$ -finite type. In other words, the set of X over S for which there is an exact sequence

$$M \longrightarrow H^{n+1}(E_X) \xrightarrow{H^{n+1}(u)} H^{n+1}(F_X), \quad M \in \text{ob } \mathcal{C}_{0,X},$$

is a refinement of S . Since $\mathcal{C}_0$ is locally quasi-liftable in $\mathcal{C}$, one may moreover impose that the arrow

$$M \longrightarrow H^{n+1}(E_X)$$

factor through $Z^{n+1}(E_X)$:

$$M \xrightarrow{r} Z^{n+1}(E_X) \longrightarrow H^{n+1}(E_X).$$

The composite

$$M \xrightarrow{u_X^{n+1}r} Z^{n+1}(F_X) \longrightarrow H^{n+1}(F_X)$$

is zero by construction, so $u_X^{n+1}r$ factors through

$$B^{n+1}(F_X) = \Im\left(F_X^n \xrightarrow{d_F^n} F_X^{n+1}\right).$$

Applying quasi-liftability once more, one may impose that $u_X^{n+1}r$ factor through F_X^n :

$$M \xrightarrow{s} F_X^n \longrightarrow B^{n+1}(F_X) \longrightarrow Z^{n+1}(F_X).$$

Then u' , defined as in (1.4.1), is plainly a morphism of complexes satisfying (i).

Let G' be the cone of $u' : E' \rightarrow F_X$. We show that $H^n(G')$ is of $\mathcal{C}_0$ -finite type. By construction there is an exact sequence

$$0 \longrightarrow E_X \xrightarrow{v} E' \longrightarrow M[-n] \longrightarrow 0,$$

with $u'v = u_X$. To compare G_X and G' , one may, for instance, form the octahedron in $K(\mathcal{C}_X)$:

$$\begin{array}{ccccc} & & G_X & \longrightarrow & M[-n+1] \\ & \nearrow^{+1} & \searrow & \nearrow & \\ E_X & \xrightarrow{v} & E' & \xrightarrow{u'} & F_X. \end{array}$$

The exact cohomology sequence of the distinguished triangle marked by the octahedron shows that

$$H^n(G_X) \longrightarrow H^n(G')$$

is an epimorphism; hence $H^n(G')$ is indeed of $\mathcal{C}_0$ -finite type.

The preceding correction therefore reduces us to the case where $H^{n+1}(u)$ is a monomorphism. We suppose this from now on. The exact cohomology sequence of $E \rightarrow F \rightarrow G$ then gives an exact sequence

$$H^n(E) \xrightarrow{H^n(u)} H^n(F) \longrightarrow H^n(G) \longrightarrow 0.$$

Since $H^n(G)$ is of $\mathcal{C}_0$ -finite type, the set of X over S for which there exists an epimorphism

$$M \longrightarrow H^n(G_X), \quad M \in \text{ob } \mathcal{C}_{0,X},$$

is, by definition, a refinement of S . Using the quasi-liftability of $\mathcal{C}_0$ in $\mathcal{C}$, one may impose that this epimorphism factor through $Z^n(F_X)$:

$$M \xrightarrow{s} Z^n(F_X) \longrightarrow H^n(F_X) \longrightarrow H^n(G_X).$$

It is clear that u' , defined as in (1.4.1) with $r = 0$, is a morphism of complexes and that

$$H^n(u') : H^n(E_X) \oplus M \longrightarrow H^n(F_X)$$

is an epimorphism. This completes the proof of (1.4.1).

Lemma 1.4.2. Let $\mathcal{C}$ be an abelian category, let $u : E \rightarrow F$ be a morphism of complexes of $\mathcal{C}$ (respectively an arrow of $D(\mathcal{C})$), and let G be the cone of u . For a given $n \in \mathbb{Z}$, the following conditions are equivalent:

- (i) $H^i(G) = 0$ for $i \geq n$;
- (ii) $H^i(u)$ is an isomorphism for $i > n$ and an epimorphism for $i = n$.

The proof is immediate from the exact cohomology sequence.

Definition 1.4.3. We shall say that u is an n -quasi-isomorphism (respectively an n -isomorphism) if u satisfies the equivalent conditions of (1.4.2).

That being so, one deduces at once from (1.4.1) the following:

Corollary 1.4.4. Let $S, \mathcal{S}, \mathcal{C}, \mathcal{C}_0$ be as in (1.4.1). Let $u : E \rightarrow F$ be an $(n+1)$ -quasi-isomorphism of complexes of $\mathcal{C}_S$, with $n \in \mathbb{Z}$ fixed, such that $H^n(G)$ is of $\mathcal{C}_0$ -finite type. Then the set of X over S for which there exist (M, r, s) as in (1.4.1) such that u' is an n -quasi-isomorphism is a refinement of S .

In pictorial terms, one may, using the finiteness condition on $H^n(G)$, locally “gain one notch,” that is, locally modify u in degree n by adding to E^n an object of $\mathcal{C}_0$ in such a way as to make u an n -quasi-isomorphism.

Corollary 1.4.5. Let $\mathcal{C}$ be an abelian category, let $n \in \mathbb{Z}$, and let $u : E \rightarrow F$ be an n -quasi-isomorphism of complexes of $\mathcal{C}$. There exists a commutative diagram in $\mathbf{C}(\mathcal{C})$

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ u \downarrow & \swarrow u' & \\ F & & \end{array}$$

where u' is a quasi-isomorphism and v is a monomorphism inducing the identity in degrees $\geq n$, that is,

$$E^i = E'^i, \quad v^i = \text{Id}(E^i) \quad \text{for } i \geq n.$$

Proof. Applying (1.4.4) in the case where S is the punctual site and $\mathcal{C}_0 = \mathcal{C}$, one obtains a commutative triangle in $\mathbf{C}(\mathcal{C})$

$$\begin{array}{ccc} E & \xrightarrow{v_{n-1}} & E'_{n-1} \\ u \downarrow & \swarrow u'_{n-1} & \\ F & & \end{array}$$

where u'_{n-1} is an $(n-1)$ -quasi-isomorphism and v_{n-1} is a monomorphism inducing the identity in degrees $\geq n$. Iterating the process, one finds, for $i \leq n-1$, a sequence of commutative triangles in $\mathbf{C}(\mathcal{C})$

$$\begin{array}{ccc} E'_i & \xrightarrow{v_{i-1}} & E'_{i-1} \\ u'_i \downarrow & \swarrow u'_{i-1} & \\ F & & \end{array}$$

where u'_i is an i -quasi-isomorphism and v_{i-1} is a monomorphism inducing the identity in degrees $\geq i$. The corollary follows by passing to the limit. More precisely, writing d'_i for the differential of E'_i , define E' by

$$E'^i = E^i, \quad d'^i = d^i \quad (i \geq n),$$

and

$$E'^i = (E'_i)^i, \quad d'^i = (d'_i)^i \quad (i < n),$$

and then define v and u' by

$$v^j = (v_i)^j, \quad u'^j = (u'_i)^j \quad (i \leq j).$$

One immediately verifies that E' , v , and u' , so defined, answer the question.

In abbreviated form: one can extend an n -quasi-isomorphism into a quasi-isomorphism “without changing anything given in degree $> n$.”

2. Pseudo-coherent complexes

2.0. In this section, S denotes a site, $\mathcal{S}$ an object of S , $\mathcal{C}$ a flat abelian S -category (1.1), and $\mathcal{C}_0$ a strictly full additive sub- S -category of $\mathcal{C}$. We assume that $\mathcal{C}_0$ is locally quasi-stable under kernels of epimorphisms (1.2) and locally quasi-liftable in $\mathcal{C}$ (1.2).

Definition 2.1. Let $X \in \text{ob } S$, let E be a complex of $\mathcal{C}_X$, and let $n \in \mathbb{Z}$. We shall say that E is *strictly $\mathcal{C}_0$ - n -pseudo-coherent* (respectively $\mathcal{C}_0$ -pseudo-coherent, respectively $\mathcal{C}_0$ -perfect) if $E^i = 0$ for i sufficiently large and $E^i \in \text{ob } \mathcal{C}_{0,S}$ for $i \geq n$ (respectively for every i , respectively for every i and $E^i = 0$ for almost all i).

When no confusion is to be feared, we shall sometimes say “strictly n -pseudo-coherent” (respectively, and so on) instead of “strictly $\mathcal{C}_0$ - n -pseudo-coherent”.

Lemma 2.1.1. Let $E \in \text{ob } \mathcal{C}(\mathcal{C}_S)$ and $n \in \mathbb{Z}$. If E is strictly n -pseudo-coherent and if $H^i(E) = 0$ for $i > n$, then $H^n(E)$ is of finite type.

Proof. The sequence

$$0 \longrightarrow Z^n(E) \longrightarrow E^n \longrightarrow E^{n+1} \longrightarrow \dots \longrightarrow 0$$

is exact, and $E^i \in \text{ob } \mathcal{C}_0$ for $i \geq n$. Since $\mathcal{C}_0$ is locally quasi-stable under kernels of epimorphisms, it follows that $Z^n(E)$ is of finite type, hence so is $H^n(E)$, which is a quotient of it.

Proposition 2.2. Let $F \in \text{ob } \mathcal{D}(\mathcal{C}_S)$ and $n \in \mathbb{Z}$. Then:

a) If F is strictly n -pseudo-coherent (relative to $\mathcal{C}_0$) and if

$$u : E \longrightarrow F$$

is a quasi-isomorphism, then the set of X over S such that there exists a quasi-isomorphism

$$v : F' \longrightarrow E_X$$

with F' strictly n -pseudo-coherent is a refinement of S .

b) The following conditions are equivalent:

(i) the set of X over S such that there exists a quasi-isomorphism

$$u : E \longrightarrow F_X$$

of $\mathcal{C}(\mathcal{C}_X)$, with E strictly n -pseudo-coherent, is a refinement of S ;

(i') the same condition as (i), but with u an isomorphism of $\mathcal{D}(\mathcal{C}_X)$;

(ii) the set of X over S such that there exists an n -quasi-isomorphism

$$u : E \longrightarrow F_X$$

of $\mathcal{C}(\mathcal{C}_X)$, with E strictly perfect, is a refinement of S ;

(ii') the same condition as (ii), but with u an n -isomorphism of $\mathcal{D}(\mathcal{C}_X)$.

Proof. For a), there exists, for i sufficiently large, an i -quasi-isomorphism

$$v_i : F'_i \longrightarrow E$$

with F'_i strictly n -pseudo-coherent (take $F'_i = 0$). We show that, if $i > n$, there exists, after refining S , an $(i - 1)$ -quasi-isomorphism

$$v_{i-1} : F'_{i-1} \longrightarrow E$$

with F'_{i-1} strictly n -pseudo-coherent. By (1.4.4), it is enough to show that $H^{i-1}(M')$, where M' is the cone of v_i , is of $\mathcal{C}_0$ -finite type. Now if M'' denotes the cone of uv_i , the hypothesis that u is a quasi-isomorphism and (1.4.2) imply that

$$H^j(M') \xrightarrow{\sim} H^j(M'') \quad \text{for all } j,$$

and

$$H^j(M') = 0 \quad \text{for } j \geq i.$$

The cone formula

$$M''^j = F_i'^{j+1} \oplus F^j$$

shows, on the other hand, that M'' is strictly $(i - 1)$ -pseudo-coherent. Hence, by (2.1.1),

$$H^{i-1}(M'') = H^{i-1}(M')$$

is of finite type, whence the assertion. It follows, by iteration and after refining S , that there is an n -quasi-isomorphism

$$v_n : F'_n \longrightarrow E$$

where F'_n is strictly n -pseudo-coherent. But, by (1.4.5), one can extend v_n into a quasi-isomorphism

$$v : F' \longrightarrow E$$

without changing anything in degrees $\geq n$; in particular F' is strictly n -pseudo-coherent. This proves a).

For b), the equivalence of (i) and (i') follows at once from a) and from the definition of isomorphisms of $D^-(\mathcal{C}_X)$. If $E \in \text{ob } D^-(\mathcal{C}_S)$ is strictly n -pseudo-coherent, there is an exact sequence

$$0 \longrightarrow E' \longrightarrow E \longrightarrow E'' \longrightarrow 0$$

where E' is strictly perfect and $E''^i = 0$ for $i \geq n$. Consequently $E' \rightarrow E$ is an n -quasi-isomorphism, which gives the implication (i) $\Rightarrow$ (ii). The implication (ii) $\Rightarrow$ (ii') is trivial, so it remains to prove (ii') $\Rightarrow$ (i). Let

$$u : E \longrightarrow F$$

be an n -isomorphism of $D^-(\mathcal{C}_S)$, with E strictly perfect. Then u is represented by a diagram in $C^-(\mathcal{C}_S)$

$$\begin{array}{ccc} & E' & \\ s \swarrow & & \searrow u' \\ E & & F, \end{array}$$

where s is a quasi-isomorphism and u' an n -quasi-isomorphism. By a), after refining S , there exists a quasi-isomorphism

$$t : E'' \longrightarrow E',$$

with E'' strictly n -pseudo-coherent. Applying (1.4.5) to $u't : E'' \rightarrow F$, we extend it to a quasi-isomorphism

$$E''' \longrightarrow F$$

with E''' strictly n -pseudo-coherent. This proves (ii') $\Rightarrow$ (i) and completes the proof of (2.2).

Definition 2.3. Under the hypotheses of (2.2), we shall say that $F \in \text{ob } D^-(\mathcal{C}_S)$ is $n\text{-}\mathcal{C}_0$ -pseudo-coherent if F satisfies the equivalent conditions b) of (2.2). We shall say that F is $\mathcal{C}_0$ -pseudo-coherent if F is $n\text{-}\mathcal{C}_0$ -pseudo-coherent for every n .

When this cannot cause confusion, we shall sometimes allow ourselves to say n -pseudo-coherent, pseudo-coherent, instead of $n\text{-}\mathcal{C}_0$ -pseudo-coherent, $\mathcal{C}_0$ -pseudo-coherent.

Remark 2.3.1. Let $F \in \text{ob } D(\mathcal{C}_S)$ be a pseudo-coherent complex. The set of X over S such that there exists a quasi-isomorphism

$$E \longrightarrow F_X$$

with E strictly pseudo-coherent (2.1) is not in general a refinement of S . However, we shall see later (2.7 a) and Exposé II) important cases in which this is so.

The proof of (2.2 a) gives the following result.

Proposition 2.4. Let $n, i \in \mathbb{Z}$, and let

$$u : E \longrightarrow F$$

be an i -quasi-isomorphism of $C(\mathcal{C}_S)$, with F n -pseudo-coherent and E strictly n -pseudo-coherent. Then, after refining S , there exists a commutative triangle in $C(\mathcal{C}_S)$

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ u \downarrow & \swarrow u' & \\ F & & \end{array}$$

where E' is strictly n -pseudo-coherent, u' is a quasi-isomorphism, and v is a monomorphism inducing the identity in degree $\geq i$.

Proposition 2.5. a) If $F \in \text{ob } D(\mathcal{C}_S)$ is n -pseudo-coherent (respectively pseudo-coherent), then $F[m]$ is $(n - m)$ -pseudo-coherent (respectively pseudo-coherent).

b) Let

$$\begin{array}{ccc} E & \xrightarrow{\quad} & F \\ & \swarrow \quad \searrow & \\ & +1 \quad G & \end{array} \quad (*)$$

be a distinguished triangle of $D(\mathcal{C}_S)$. If E and G are n -pseudo-coherent (respectively pseudo-coherent), then F is n -pseudo-coherent (respectively pseudo-coherent).

Proof. a) is trivial from the definitions.

b) It is enough, of course, to prove the non-parenthesized assertion. The triangle (*) can also be written

$$\begin{array}{ccc} G[-1] & \xrightarrow{\quad} & E \\ & \swarrow \quad \searrow & \\ & +1 \quad F & \end{array}$$

By a), $G[-1]$ is $(n + 1)$ -pseudo-coherent. Thus we are reduced to proving that, if E is $(n + 1)$ -pseudo-coherent and F is n -pseudo-coherent, then G is n -pseudo-coherent. After localizing, we

may assume E (respectively F) strictly $(n+1)$ -pseudo-coherent (respectively n -pseudo-coherent). The arrow

$$E \xrightarrow{u} F$$

of $D^-(\mathcal{C}_S)$ is represented by a pair of arrows of $C^-(\mathcal{C}_S)$

$$\begin{array}{ccc} & E' & \\ s \swarrow & & \searrow u' \\ E & & F, \end{array}$$

where s is a quasi-isomorphism. By (2.2 a)), after localizing, there exists a quasi-isomorphism

$$t : E'' \longrightarrow E',$$

where E'' is strictly $(n+1)$ -pseudo-coherent. We are therefore reduced to the case where u is defined by a true arrow of $C^-(\mathcal{C}_S)$. The assertion then follows from the definitions and from the formula

$$C(u)^i = E^{i+1} \oplus F^i,$$

where $C(u)$ denotes the cone of u .

Scholium 2.6. By rotation of triangles, one sees that pseudo-coherence properties for two of the objects of the triangle $(*)$ imply another such property for the third. The various cases may be summarized in the table below; the lower-case letters denote degrees of pseudo-coherence, and in each row the conclusion is underlined:

E	F	G
$n+1$	n	<u>n</u>
n	<u>n</u>	n
<u>n</u>	n	$n-1$.

We shall write

$$D(\mathcal{C})_{\mathcal{C}_0\text{-}n\text{coh}} \quad \text{respectively} \quad D(\mathcal{C})_{\mathcal{C}_0\text{-}coh}$$

for the strictly full fibered subcategory of $D(\mathcal{C})$ whose fiber over each $X \in \text{ob } S$ is formed by the objects of $D(\mathcal{C}_X)$ that are n -pseudo-coherent (respectively pseudo-coherent) relative to $\mathcal{C}_0$. By (2.5), the category

$$D(\mathcal{C})_{\mathcal{C}_0\text{-}coh}$$

is a triangulated sub- S -category of $D(\mathcal{C})$ (1.1).

Proposition 2.7. Suppose that S is the punctual site, so that

$$\mathcal{C} = \mathcal{C}_S, \quad \mathcal{C}_0 = \mathcal{C}_{0,S}.$$

Then:

- a) If $F \in \text{ob } D(\mathcal{C})$ is pseudo-coherent, there exists a quasi-isomorphism

$$E \longrightarrow F$$

with E strictly pseudo-coherent (2.1).

- b) Let $K^{-,ac}(\mathcal{C}_0)$ denote the full triangulated subcategory of $K^-(\mathcal{C}_0)$ formed by acyclic complexes. The canonical functor

$$K^-(\mathcal{C}_0)/K^{-,ac}(\mathcal{C}_0) \longrightarrow D(\mathcal{C})$$

is fully faithful and has essential image $D(\mathcal{C})_{\mathcal{C}_0\text{-}coh}$.

Proof. a) Let $F \in \text{ob } D(\mathcal{C})_{\mathcal{C}_0\text{-coh}}$, and suppose constructed an i -quasi-isomorphism

$$u_i : E_i \longrightarrow F$$

of $C(\mathcal{C})$, with E_i strictly pseudo-coherent; this is possible for i sufficiently large, taking $E_i = 0$. Let G_i be the cone of u_i . It follows from (1.4.2) and (2.1.1) that $H^{i-1}(G_i)$ is of $\mathcal{C}_0$ -finite type. Hence, by (1.4.4), there exists a commutative triangle in $C(\mathcal{C})$

$$\begin{array}{ccc} E_i & \xrightarrow{v_{i-1}} & E_{i-1} \\ u_i \downarrow & \swarrow u_{i-1} & \\ F & & \end{array}$$

where E_{i-1} is strictly pseudo-coherent, u_{i-1} an $(i-1)$ -quasi-isomorphism, and v_{i-1} a monomorphism inducing the identity in degrees $\geq i$. Passing to the limit (cf. the proof of (1.4.5)) gives a quasi-isomorphism

$$u : E \longrightarrow F$$

with E strictly pseudo-coherent.

b) Apply the criterion of [V], Chap. I, §2, 4.2 b)(i), together with a).

We note that the proof of a) gives the following more precise result, which generalizes (1.4.5).

Proposition 2.7.1. Under the hypotheses of (2.7), let $u : E \rightarrow F$ be an n -quasi-isomorphism of $C(\mathcal{C})$, with F pseudo-coherent and E strictly pseudo-coherent. Then there exists a commutative triangle in $C(\mathcal{C})$

$$\begin{array}{ccc} E & \xrightarrow{v} & E' \\ u \downarrow & \swarrow u' & \\ F & & \end{array}$$

where E' is strictly pseudo-coherent, u' is a quasi-isomorphism, and v is a monomorphism inducing the identity in degree $\geq n$.

Definition 2.8. Let $n \in \mathbb{N}$ and $F \in \text{ob } \mathcal{C}_S$. We shall say that F is of n - $\mathcal{C}_0$ -finite presentation if the set of X over S such that there exists an exact sequence

$$E^{-n} \longrightarrow \cdots \longrightarrow E^0 \longrightarrow F_X \longrightarrow 0, \quad E^i \in \text{ob } \mathcal{C}_{0,X} \quad (-n \leq i \leq 0),$$

is a refinement of S . We shall say that F is of ∞ - $\mathcal{C}_0$ -finite presentation if F is of n - $\mathcal{C}_0$ -finite presentation for every $n \in \mathbb{N}$.

Of course, we shall sometimes simply say n -finite presentation when there is no possible ambiguity about $\mathcal{C}_0$. We shall say “finite presentation” for “1-finite presentation,” and “finite type” for “0-finite presentation,” the latter notion coinciding with the one already introduced in (1.2).

Proposition 2.9. Let $n \in \mathbb{N}$ and $F \in \text{ob } \mathcal{C}_S$. The following conditions are equivalent:

- (i) F is of n -finite presentation (respectively of ∞ -finite presentation);
- (ii) F is $(-n)$ -pseudo-coherent (respectively pseudo-coherent).

In the case where S is the punctual site, F is pseudo-coherent if and only if there exists an exact sequence

$$\cdots \longrightarrow E^i \longrightarrow \cdots \longrightarrow E^0 \longrightarrow F \longrightarrow 0, \quad E^i \in \text{ob } \mathcal{C}_0 \quad \text{for all } i.$$

Proof. For the equivalence (i) $\Leftrightarrow$ (ii), it is plainly enough to treat the non-parenthesized case. The implication (i) $\Rightarrow$ (ii) is immediate from the definitions, and (ii) $\Rightarrow$ (i) follows at once from

(2.4): the arrow $0 \rightarrow F$ is a 1-quasi-isomorphism, and from this one deduces, after localizing, a quasi-isomorphism

$$E \longrightarrow F$$

with E strictly $(-n)$ -pseudo-coherent and $E^i = 0$ for $i \geq 1$, that is, an n -finite presentation of F . The second assertion is proved in the same way, using (2.7.1).

Proposition 2.10. a) Let $m, n \in \mathbb{Z}$, with $n \leq m$. If $F \in \text{ob } D^-(\mathcal{C}_S)$ is n -pseudo-coherent and acyclic in degrees $\geq m$, then there exists locally a quasi-isomorphism

$$E \longrightarrow F$$

with E strictly n -pseudo-coherent and $E^i = 0$ for $i \geq m$. If S is punctual, and if F is pseudo-coherent and acyclic in degrees $\geq m$, there exists a quasi-isomorphism

$$E \longrightarrow F$$

with E strictly pseudo-coherent and $E^i = 0$ for $i \geq m$.

b) Let $F \in \text{ob } D^-(\mathcal{C}_S)$ be acyclic in degrees $> n$. For F to be n -pseudo-coherent, it is necessary and sufficient that $H^n(F)$ be of finite type.

Proof. a) Since F is acyclic in degrees $\geq m$, the arrow $0 \rightarrow F$ is an m -quasi-isomorphism. The assertion follows from (2.4) and (2.7.1).

b) There exists a distinguished triangle in $D^-(\mathcal{C}_S)$:

$$\begin{array}{ccc} F & \xrightarrow{\quad} & F' \\ & \swarrow \scriptstyle +1 & \searrow \\ & H^n(F)[-n] & \end{array}$$

(to see this, begin by truncating F , replacing F^n by $Z^n(F)$ and F^i by 0 for $i > n$). The exact cohomology sequence implies

$$H^i(F') = 0 \quad \text{for } i \geq n - 1,$$

so F' is $(n - 1)$ -pseudo-coherent. We conclude by (2.5) and (2.6).